

Privacy, Reliability, Transparency, and Proportionality in Artificial Intelligence Regulation

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Abstract

Artificial intelligence (AI) systems increasingly mediate decisions affecting individuals and societies. Existing data protection frameworks address certain privacy-related harms, particularly those arising from data leakage, re-identification, and profiling. However, they inadequately capture a more fundamental risk: unreliable or unjustified inference produced by AI systems even when data collection and processing are legitimate. This article argues that modern AI raises distinct concerns of construct validity, confounding, representativeness, distribution shift, and fairness trade-offs that require specialised regulatory attention. In the context of AI, moreover, transparency and explainability acquire distinct and significantly more challenging meanings. Effective governance must therefore incorporate AI-specific threat modelling, post-deployment monitoring, operational guidelines for legitimate use, and proportionality assessments grounded in structured articulation of both risks and benefits.

1 Introduction

AI systems are rapidly becoming embedded in fields such as finance, education, employment, healthcare, policing, and public administration. They rely on large-scale data collection and model-based inference to make predictions or classifications about individuals. While policy attention has focused heavily on privacy and data protection, this can obscure a deeper challenge: the reliability and fairness of the inferences themselves.

Traditional privacy regulation assumes that harm arises primarily from unauthorized access, disclosure, or misuse of personal data. Yet AI systems can cause significant harm even when data is lawfully collected, securely stored, and processed for approved purposes. The central issue is therefore not only how data is handled, but whether the conclusions drawn from it are justified.

Recent work in the epistemology and engineering of machine learning emphasizes that strong performance on historical datasets does not guarantee reliable real-world behaviour, particularly under changing conditions or novel inputs [12, 15, 28, 25].

2 Existing Data Protection Frameworks and AI

Two categories of risk are relatively well addressed within current data protection paradigms.

Data leakage: Unauthorized disclosure of personal information remains a core privacy concern. Legal and technical safeguards such as access control, encryption, and privacy-preserving federated computation aim to prevent such leakage. While implementing and *verifying* these protections in complex socio-technical systems with certifiable guarantees is technically demanding, the underlying regulatory principles are well established.

Profiling and re-identification: AI systems can combine disparate datasets to infer identities or sensitive attributes, even from anonymized data. The biggest risks arise from profiling individuals by breaking informational silos. Data protection regimes typically address these risks through purpose limitation, data minimization, and restrictions on secondary use. Again, verifiable enforcement is non-trivial, in part because inference and linkage can be indirect, but the conceptual framework is clear.

A more serious regulatory gap concerns harm arising from incorrect, unstable, or unjustified inference. Existing frameworks generally do not evaluate whether AI predictions are epistemically sound or socially defensible.

Modern machine learning often learns patterns from historical correlations rather than causal relationships, making it vulnerable to spurious associations and instability under new conditions [26, 31]. This is not merely a technical problem: unreliable inference can directly undermine fairness, due process, and democratic legitimacy.

3 Validity and reliability of AI inference

The reliability of AI and data-driven systems ultimately depends on the validity of the predictions or inferences they produce. Unlike conventional software, whose correctness can often be checked against explicit rules or specifications, AI systems derive predictions from associations learned from data. Evaluating reliability therefore requires asking whether those inferences validly represent the phenomena they are intended to capture.

Three familiar dimensions of validity are especially important. Construct validity concerns whether the variables and models truly capture the concepts they purport to measure. Internal validity asks whether the inferred relationships reflect underlying mechanisms rather than artefacts of data collection, sampling, or model design. External validity asks whether conclusions drawn from one dataset or deployment setting generalize reliably to others. Together, these determine whether data-driven inference can be considered dependable in real-world decision-making.

3.1 Construct Validity in AI and Data Science

Construct validity concerns how accurately an operationalized variable represents an underlying abstract concept or “construct.” In many AI and Data Science applications, the attributes we seek to predict or infer — such as “creditworthiness,” “health status,” or “intent” — are not directly observable. AI systems therefore rely on proxy variables extracted from observational data, raising the possibility that predictions reflect measurement artefacts rather than the target phenomenon. Examples include:

- Creditworthiness is inferred from correlations between variables such as income, repayment history, and residential stability and population-level outcomes, rather than from a direct assessment of the individual’s financial prudence.
- Job suitability is inferred statistically from variables such as educational background and employment history by comparison with population-level data, rather than from a direct assessment of the individual’s creativity, competence, or integrity.
- Educational ability is inferred from standardized test scores.

Such proxies may correlate with outcomes without representing the underlying trait. If C represents the latent construct of interest and M the metric used to capture it, construct validity is the degree to which M is a high-fidelity reflection of C . In a simple linear latent-variable model,

$$M = \alpha + \beta C + \epsilon$$

where α represents systematic measurement bias, β the structural relationship between the measurement and the construct, and ϵ measurement error. Construct validity is high when β is large and ϵ is small and non-systematic.

The absence of construct validity undermines not only fairness but also technical **reliability**. If a model captures noise (ϵ) rather than the signal (C), its performance becomes fragile. The resulting harm undermines informational self-determination, a core dimension of privacy that acknowledges that harm arises not only from disclosure, but from inference [1].

3.1.1 Spurious Correlations and Model Fragility

When a model lacks construct validity, it may optimize for proximate artefacts rather than the target construct. A computer vision model trained to identify “pneumonia,” for example, may instead learn the presence of portable X-ray metal tokens used in particular hospital wards, artefacts of particular X-ray machines [35], or even whether the patient is imaged in the supine or erect position (see Figure ??). Such a model may show high accuracy on internal benchmarks yet fail when deployed in settings where these artefacts are absent.

Text models likewise inherit gender-profession biases from training corpora, and multimodal systems can learn cross-cultural text-image associations that fail in different contexts. These cases illustrate the need for causal analysis and debiasing if AI systems are to be robust.

3.1.2 Non-identifiability of latent constructs

A fundamental limitation of construct validity in data-driven systems arises from the non-identifiability of latent variables. Let X denote observable variables and Z a latent construct of interest. In general, the marginal distribution $P(X)$ does not uniquely determine the joint distribution $P(X, Z)$, and hence does not identify the conditional $P(Z | X)$. There are infinitely many joint distributions $P(X, Z)$ consistent with the same $P(X)$, so multiple, mutually incompatible latent explanations can generate the same observable data. This is a manifestation of the broader identification problem in statistics and econometrics [23, 26].

Consequently, even with unlimited data, inference about Z from X alone is underdetermined unless supplemented by strong structural assumptions, causal models, or external information. Predictive accuracy over observed outcomes therefore does not show that the inferred construct is meaningful or correctly identified. This limitation is especially acute for behavioural or social attributes — such as preferences, intent, or creditworthiness — where different latent mechanisms can generate indistinguishable patterns of observed behaviour. From a measurement-theoretic perspective, this is a failure of construct validity: the observables do not uniquely operationalize the underlying construct [7]. Any interpretation of such inferred constructs must therefore be treated as contingent on unverified modelling assumptions rather than as empirically established fact.

3.1.3 Omitted Variable Bias (OVB)

In any complex domain, what is measured is only a small fraction of what matters. The overwhelming majority of causally relevant factors remain unobserved. Examples include:

- Motivation and ethical judgment in hiring decisions
- Informal income or family support in credit risk
- Genetic mutations or environmental exposures in health outcomes

No algorithm can correct for variables that were never measured. When essential variables are absent, models rely on correlated substitutes, producing unreliable individual predictions.

If the true underlying construct C is influenced by multiple factors $X_1, X_2, \dots, X_n$, omitting a key factor X_i that is correlated with both the input and the target leads to

omitted variable bias. In a regression setting, the bias in the estimated coefficient $\hat{\beta}_1$ for an included variable X_1 when a relevant variable X_2 is omitted is

$$\text{Bias}(\hat{\beta}_1) = \beta_2 \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$$

where β_2 is the causal effect of the omitted variable. The resulting model reflects a distorted version of reality, affecting both fairness and reliability. For example, predicting “customer loyalty” using only “purchase frequency” while omitting “market competition” creates a construct gap that leads to unreliable churn forecasts [20].

3.1.4 Label bias

Supervised learning assumes that historical labels represent ground truth. In many domains, however, labels reflect institutional decisions rather than objective outcomes [2]. Examples include:

- Arrest records reflecting policing patterns rather than crime rates
- Hiring data reflecting past discrimination
- Loan approvals reflecting institutional risk tolerance

Models trained on such labels may therefore reproduce historical behaviour rather than infer the intended construct.

3.1.5 Aggregation effects

Population-level relationships may not hold for individuals, leading to ecological fallacies. For example, neighbourhood characteristics may be used to infer individual socioeconomic status or risk.

3.1.6 Rare events

Many important outcomes depend on rare or unmeasured factors, such as financial crises or extreme weather events. Models optimized for average performance may therefore fail precisely where stakes are highest.

3.1.7 Feedback effects and performativity

AI predictions can influence the very outcomes they seek to predict. For example:

- Predictive policing directing resources to specific areas, generating more recorded crime there
- Credit scores influencing borrowing behavior

- Risk assessments affecting sentencing or parole decisions

Such feedback loops can amplify biases and degrade validity over time [9]. When individuals adapt their behaviour to optimize algorithmic scores, the proxy becomes detached from the construct. This is closely related to Goodhart’s law: when a measure becomes a target, it ceases to be a good measure [11, 33].

3.2 Internal validity and confounding

Internal validity concerns whether observed relationships reflect causal connections rather than spurious correlations. A central difficulty in evaluating AI reliability is *confounding*: a variable may influence both observed inputs and target outputs, inducing correlations that do not reflect the intended causal relationship [27]. When machine learning systems are trained purely on observational data, such spurious correlations can easily be absorbed into the learned model.

Examples include:

- Socioeconomic status affecting both education and health outcomes
- Institutional practices influencing both hiring decisions and employee performance
- Environmental factors affecting both location and disease incidence

Failure to account for confounders can produce systematically biased predictions.

Large Language Models (LLMs) are especially susceptible because they are trained on massive corpora drawn from heterogeneous internet sources. These corpora contain strong statistical regularities that arise from historical, social, or procedural artefacts rather than the semantic structure of the underlying tasks. When an LLM is trained to approximate conditional distributions of the form

$$P(y \mid x),$$

it has no intrinsic mechanism to distinguish whether predictive signal arises from the intended semantic relation between x and y , or from some latent confounding variable z that simultaneously influences both.

Empirical studies have repeatedly shown that models trained on popular NLP benchmarks exploit such confounding signals. In Natural Language Inference datasets such as SNLI and MultiNLI, annotation artefacts allow models to predict labels using only the hypothesis sentence without examining the premise [13]. Similarly, question-answering systems trained on datasets such as SQuAD often exploit superficial lexical overlap between the question and candidate answer sentences rather than performing genuine semantic reasoning [17].

More broadly, language models internalise statistical associations present in training corpora, including demographic stereotypes and other social regularities, which may

function as latent confounders influencing predictions [4, 6]. Such effects show that high benchmark accuracy does not imply that a model has learned the intended conceptual or causal structure of the task.

From a reliability perspective, confounding is a fundamental challenge. If a model’s predictions depend on spurious correlations rather than the true semantic relationship underlying the task, its performance may degrade sharply when deployed in environments where those correlations no longer hold. This phenomenon, often described as *dataset artefacts* or *shortcut learning*, has been widely observed across modern machine learning systems [10]. Addressing confounding therefore requires not only larger datasets, but also better methods for causal analysis, dataset construction, and robustness evaluation.

Causal inference has long emphasized that reliable generalization requires reasoning about mechanisms and confounding [26, 16]. Historical data also often embeds social and institutional bias, compounding the risk of systematically misleading outputs [2].

3.3 External validity and distribution shift

Another major challenge in assessing AI reliability concerns *external validity*: the extent to which conclusions derived from a particular dataset or setting remain valid when the system is deployed elsewhere [32]. In machine learning, this problem arises because models are trained on specific datasets with specific distributions but are often deployed in environments where the underlying distribution differs. Such distribution shift can result from demographic differences, temporal changes, policy or behavioural adaptations, measurement changes, and similar factors.

Formally, most machine learning systems are trained to minimise an expected loss under a training distribution $P_{\text{train}}(x, y)$. During deployment, however, the system may encounter data drawn from a different distribution $P_{\text{test}}(x, y)$. When

$$P_{\text{train}}(x, y) \neq P_{\text{test}}(x, y),$$

performance guarantees obtained during training may no longer hold. This is the problem of *distribution shift* [29].

Several forms of distribution shift are well documented. In computer vision, models trained on curated datasets such as ImageNet show significant accuracy degradation when evaluated on newly collected but semantically similar images, illustrating sensitivity to subtle changes in data collection procedures [30]. Benchmark datasets also often contain hidden correlations between labels and background features; when these correlations change, models that relied on them exhibit sharp performance drops [34].

In natural language processing, distribution shifts arise across domains, genres, and time periods. Models trained on newswire or web text often perform poorly when applied to social media, conversational dialogue, or specialised domains such as medicine or law [3]. Even within similar domains, linguistic usage evolves over time, producing temporal

shifts that degrade the performance of language models trained on earlier corpora [19].

Large Language Models partially mitigate some of these effects through large-scale pretraining across diverse sources, but they remain susceptible to shift in downstream applications. When prompts, task formulations, or domain contexts differ from those represented in training data, model behaviour can change unpredictably. Empirical studies document substantial variability in LLM performance across prompt formulations, task domains, and evaluation settings [21].

From a reliability perspective, distribution shift directly undermines external validity. High benchmark performance therefore cannot be taken as evidence of reliable operational performance. Addressing this challenge requires systematic evaluation across diverse data distributions, together with methods for domain adaptation, robustness testing, and continual learning. In some cases, there may not even be an identifiable probability distribution over the input space because there is no well-defined universe. Without very specific context, for example, the probability of the next prompt to a language model is not even a meaningful concept. In such situations, the lack of a theoretical framework makes reliability analysis even more difficult.

Dataset shift is widely recognized as a central challenge in machine learning [28]. Empirical studies also show that predictive uncertainty estimates can fail dramatically under such shifts [25].

4 Bias, fairness, and the limits of statistical correction

Concerns about bias and fairness have become central to AI governance. A large body of work proposes formal definitions of fairness — such as demographic parity, equalized odds, and calibration — together with algorithmic techniques to mitigate disparities [14, 18]. These are useful diagnostic tools, but they do not fully address the deeper sources of unfairness in data-driven systems.

Bias in AI is often not merely a statistical artifact but a consequence of more fundamental validity failures, including construct ambiguity, confounding, ecological bias, and non-identifiability of latent variables.

4.1 Sources of bias beyond data imbalance

Bias is commonly attributed to imbalanced or unrepresentative datasets. However, even perfectly balanced data may yield unfair outcomes when the underlying inference problem is ill-posed.

Construct bias When the target variable is only a proxy for the intended construct, predictions may systematically disadvantage certain groups. Using healthcare expendi-

ture as a proxy for health need, for example, can underestimate the needs of populations with limited access to care [24]. This is a failure of construct validity rather than a simple data imbalance.

Confounding and structural bias Historical data often reflects structural inequalities and institutional practices. Machine learning models trained on such data may reproduce these patterns even when they are causally irrelevant to the intended outcome [2]. Unobserved confounders can further distort relationships.

Ecological bias Many AI systems apply population-level correlations to individuals. This ecological fallacy can produce systematic misclassification, particularly for individuals whose characteristics deviate from group averages. Credit scoring and targeted advertising are especially susceptible.

Non-identifiability of fairness-relevant constructs As noted earlier, latent constructs such as risk, preference, or intent are often not identifiable from observable data. Different latent structures may produce identical observable distributions, making it impossible to determine whether disparities reflect true differences or modelling artefacts. This limits the extent to which fairness can be assessed or corrected using observational data alone.

4.2 Incompatibility of fairness criteria

A central result in the fairness literature is that different statistical definitions of fairness cannot, in general, be satisfied simultaneously [18]. It is typically impossible, for example, to achieve both calibration and equal error rates across groups when base rates differ.

This incompatibility shows that fairness is not a purely technical property but requires normative choices about which criteria to prioritize. Algorithmic methods cannot resolve these trade-offs on their own.

4.3 Limits of algorithmic mitigation

Techniques such as reweighting, fairness constraints, or post-processing adjustments aim to reduce disparities in model outputs. While useful, they have important limitations:

- They operate on observed variables and cannot correct for unobserved confounders.
- They assume that the target construct is valid and well-defined.
- They may introduce new distortions or trade-offs across groups.
- They do not address feedback effects or behaviour induced by the system itself.

As a result, algorithmic fairness interventions may improve statistical metrics without addressing deeper sources of unfairness.

4.4 Fairness as a validity problem

Fairness is closely linked to validity:

- If the construct being predicted is ill-defined, fairness with respect to that construct is ill-defined.
- If predictions are confounded or non-identifiable, observed disparities may not reflect meaningful differences.
- If models rely on ecological correlations, individual-level fairness cannot be guaranteed.

Fairness therefore cannot be ensured independently of validity analysis.

5 Summary of Impacts

The technical and social consequences of poor construct validity are summarized in Table 1.

Dimension	Technical Consequence	Real-World Failure
Fairness	Biased Proxies	Systematic exclusion of marginalized groups.
Reliability	Sensitivity to Noise (ϵ)	Model performance collapses in production.
Interpretability	Omitted Variable Bias	Decisions based on false causal links.
Robustness	Distribution Shift Failure	AI fails when sensor types or contexts change.

Table 1: Impacts of Construct Validity Failure

6 Transparency and Explainability in AI Systems

Transparency has long been regarded as a central principle in the governance of computational systems. In traditional software, apart from clarity about design and deployment processes, transparency typically refers to the ability to inspect and understand the logic of the system. Because such systems are built from explicitly specified rules and procedures, examining the source code can, in principle, reveal how outputs are produced. Errors or unintended behaviour can often be traced to identifiable components of the program.

In modern AI systems, particularly those based on machine learning, the meaning of transparency changes significantly. Models such as deep neural networks and large language models are not defined by explicit logical rules but are learned from data through large-scale optimisation. Their behaviour is encoded in high-dimensional parameter

spaces consisting of millions or billions of numerical weights. Although the architecture and parameters may be fully accessible, that information rarely provides a clear account of how a specific prediction or decision is produced [22]. Classical transparency—simply revealing internal mechanisms—therefore does not necessarily yield meaningful understanding.

Closely related is the notion of *explainability*, which concerns the ability of a system to provide human-interpretable reasons for its outputs. Unlike transparency, which focuses on access to internal structure, explainability aims to produce explanations that are understandable and useful to users or auditors [8]. These may take the form of feature attributions, example-based justifications, or simplified surrogate models approximating the behaviour of more complex systems.

In the context of large language models, explainability poses particular challenges. LLMs are trained to model conditional probability distributions over token sequences, and their predictions arise from complex interactions within high-dimensional representation spaces learned during training. These internal representations often do not correspond to clearly interpretable semantic concepts, making it difficult to produce explanations that faithfully reflect the underlying decision process [5]. Moreover, deep learning systems, including LLMs, typically exhibit significant redundancy in their internal representations: the same information is distributed and encoded across multiple neurons, layers, and pathways. As a result, there is no unique or canonical representation of a concept within the model, and different internal configurations may support the same output. This redundancy further undermines post-hoc interpretability, since explanations based on specific features, activations, or attribution scores may capture only one of many equivalent representations rather than the actual basis of the model’s behaviour. Post-hoc interpretability methods may therefore provide useful intuition, but they do not necessarily reveal the true internal basis of the model’s outputs.

More fundamentally, many AI systems are designed precisely to discover complex statistical associations in data that cannot be readily verbalised or expressed in logical or symbolic form. Demanding post hoc explanations in such settings may therefore be conceptually misplaced: the system’s utility often lies in capturing patterns that resist human articulation. Attempts to extract simplified explanations may obscure rather than clarify the basis of the model’s behaviour.

These considerations suggest that transparency in AI cannot be reduced to access to code, parameters, or post hoc explanations. Instead, it must be understood more broadly to include systematic evaluation of what the system is actually learning and whether its inferences are valid. In particular, meaningful transparency must incorporate analysis of construct validity, confounding, and robustness under distribution shift. Without such validity analysis, explanations—however intuitive—may simply rationalize outputs whose underlying basis is unreliable or unjustified.

This broader conception of transparency is also essential for proportionality analy-

sis. Proportionality requires that the benefits of an AI system be weighed against its potential harms. However, such an assessment presupposes that the system’s outputs constitute reliable and meaningful inferences. Where validity is uncertain or absent, neither benefits nor harms can be properly characterised, rendering proportionality analysis ill-defined. Transparency, understood as visibility into the validity and limitations of inference, therefore becomes a precondition for meaningful proportionality assessment.

It also connects directly to the constitutional principle of informational self-determination articulated in *Puttaswamy*. That principle recognises the individual’s right to control how information about them is collected, used, and interpreted. In the context of AI, this extends beyond data collection to the legitimacy of the inferences drawn from that data. Subjecting individuals to opaque or invalid inferences undermines their ability to understand, contest, or control how they are represented within algorithmic systems. Transparency, in this deeper sense, is therefore not merely a technical requirement but a condition for preserving autonomy and informational self-determination.

Effective oversight of AI systems thus depends not only on disclosure of technical artefacts such as model architectures, training data, or weights, but also on rigorous methods for interpreting, evaluating, and auditing system behaviour across a range of scenarios. Transparency, in this broader sense, is inseparable from the question of whether the system’s inferences are epistemically defensible.

7 A validity-grounded framework for risk articulation and regulation

Existing regulatory approaches to AI systems often classify risk by application domain (e.g., healthcare, law enforcement) or potential impact. While useful, such classifications do not directly address a more basic determinant of harm: whether the inference performed by the system is epistemically well-founded.

This section proposes a framework for risk mitigation grounded in *validity of inference*. The central premise is that regulatory scrutiny should be proportional not only to the consequences of deployment but also to the extent to which the target construct is meaningfully inferable from available data.

7.1 Validity as a precondition for proportionality

Proportionality analysis requires balancing benefits against harms. But if the underlying inference lacks validity, such balancing becomes meaningless. An unreliable system cannot be justified by its potential benefits, because its outputs do not constitute reliable knowledge.

Validity should therefore be treated as a *precondition* for legitimate use. Only when an inference is epistemically defensible does it become meaningful to evaluate proportionality.

7.2 From domain-based risk to epistemic risk

An AI system may be deployed in a low-impact domain yet still produce systematically misleading inferences, or operate in a high-impact domain with well-understood and reliable measurements. Domain-based classification is therefore incomplete.

Risk should instead be evaluated along dimensions of *epistemic reliability*, including:

- **Construct validity:** Is the target phenomenon meaningfully inferable from the observed data?
- **Causal adequacy:** Are relationships stable and robust to confounding?
- **Contextual stability:** Do predictions generalize across time, populations, and settings?
- **Endogeneity:** Does the system influence the data-generating process itself?
- **Fairness:** Are fairness assessments based on scrutiny of the construct being inferred, rather than only on model outputs?

These dimensions determine whether the outputs of the system can be interpreted as knowledge rather than mere correlation.

7.3 Classes of inference and their validity limits

Different classes of inference present different validity challenges. We distinguish four broad categories:

Physical systems In domains such as weather prediction or engineering diagnostics, the underlying processes are governed by physical laws. While observations may be incomplete and systems chaotic, the target construct is often well-defined and, in principle, inferable given sufficient information.

Regulatory stance: Emphasis on uncertainty quantification, robustness to rare events, and integration with physical models.

Biological systems Medical diagnosis and prognosis involve complex, partially observed systems with high variability across individuals. While constructs such as disease states are meaningful, they are affected by unobserved variables and evolving knowledge.

Regulatory stance: Strong requirements for clinical validation, population diversity, uncertainty estimation, and post-deployment monitoring.

Behavioural systems Systems such as recommendation engines, credit scoring, and targeted advertising infer latent attributes (preferences, risk, intent) from observed behavior. These constructs are:

- Context-dependent and temporally unstable
- Influenced by platform design and external conditions
- Subject to feedback loops and strategic behavior

Observed actions do not reveal stable underlying traits but conditional responses to environment and incentives. Such systems also often rely on population-level correlations to make individual-level predictions, leading to ecological bias.

Regulatory stance: These systems should be treated as inherently high validity risk. Their use in consequential decision-making should be restricted and, where permitted, accompanied by strong safeguards, including transparency, user control, and limits on personalization.

Normative constructs Certain applications attempt to infer inherently subjective or socially constructed attributes, such as trustworthiness, fairness, or “fit.” These constructs lack stable, objective definitions and cannot be reliably inferred from observational data.

Regulatory stance: Strong presumption against automated inference. Decisions should rely on human judgment, procedural safeguards, and deliberative processes rather than statistical prediction.

7.4 Feedback, performativity, and instability

Across behavioural and normative domains, AI systems often exhibit *performativity*: predictions influence outcomes, which then reshape future data. This leads to:

- Reinforcement of existing patterns
- Suppression of unobserved alternatives
- Increasing divergence between model predictions and underlying reality

Such dynamics undermine both construct and external validity over time. Regulatory frameworks must therefore treat these systems as dynamic interventions rather than passive predictors.

7.5 Fairness and bias

Bias in AI systems is often a symptom of deeper epistemic limitations rather than merely a problem of data imbalance or algorithm design. Addressing fairness therefore requires moving beyond statistical adjustments toward a more fundamental evaluation of inference validity. Without such grounding, efforts to ensure fairness risk masking arbitrariness rather than eliminating it.

A validity-centered view of fairness has several implications for AI governance:

- Fairness assessments should begin with scrutiny of the construct being inferred, not just model outputs.
- Regulatory frameworks should require analysis of confounding, omitted variables, and ecological bias.
- In high-validity-risk settings, statistical fairness metrics should not be treated as sufficient evidence of fairness.
- Normative decisions about fairness criteria should be made explicitly and subject to public scrutiny.

Where constructs are not reliably inferable, fairness concerns may justify restricting or prohibiting automated decision-making altogether.

7.6 Mitigation strategies aligned with validity

Risk mitigation should be aligned with the specific validity challenges of each class of system:

For construct validity

- Require explicit definition of the target construct
- Justify its inferability from available data
- Identify omitted variables and measurement limitations

For internal validity

- Analyse potential confounders and causal assumptions
- Use domain knowledge to assess plausibility of relationships

For external validity

- Conduct evaluation across diverse populations and conditions
- Monitor performance under distribution shift

For behavioural and feedback systems

- Assess context dependence and temporal instability
- Analyse feedback loops and performative effects
- Limit use in high-stakes individual decision-making

7.7 Graduated regulatory obligations

Rather than rigid tiers, the framework supports *graduated obligations*:

- Low validity risk: standard testing and documentation
- Moderate validity risk: enhanced evaluation and transparency
- High validity risk: restricted use, mandatory oversight, and continuous monitoring
- Non-inferable constructs: prohibition or strict limitation

The level of obligation should increase with the degree of epistemic uncertainty and potential for systematic error.

8 Conclusion

Existing data protection regimes address risks related to data leakage and re-identification but are insufficient for governing AI systems whose primary dangers arise from unreliable or unjustified inference. Construct validity, confounding, representativeness, distribution shift, tail-risk behaviour, and fairness trade-offs are intrinsic to statistical learning and require AI-specific regulatory approaches.

Effective governance must incorporate rigorous threat modelling, post-deployment monitoring, operational tests for legitimate use, and proportionality assessments grounded in explicit articulation of risks and benefits.

References

- [1] Justice K. S. Puttaswamy (Retd.) and Others v. Union of India and Others. (2017) 10 SCC 1; AIR 2017 SC 4161, 2017. Supreme Court of India, Writ Petition (Civil) No. 494 of 2012, decided 24 August 2017.
- [2] Solon Barocas and Andrew D. Selbst. Big data’s disparate impact. *California Law Review*, 104(3):671–732, 2016. URL <https://www.cs.yale.edu/homes/jf/BarocasSelbst.pdf>. Open-access author PDF.

- [3] John Blitzer, Mark Dredze, and Fernando Pereira. Biographies, bollywood, boom-boxes and blenders: Domain adaptation for sentiment classification. In *Proceedings of ACL*, 2007.
- [4] Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. In *Advances in Neural Information Processing Systems*, 2016.
- [5] Rishi Bommasani, Drew Hudson, Ehsan Adeli, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [6] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 2020.
- [7] Lee J. Cronbach and Paul E. Meehl. Construct validity in psychological tests. *Psychological Bulletin*, 52(4):281–302, 1955. doi: 10.1037/h0040957. URL <https://doi.org/10.1037/h0040957>.
- [8] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*, 2017.
- [9] Danielle Ensign, Sorelle A. Friedler, Scott Neville, Carlos Scheidegger, and Suresh Venkatasubramanian. Runaway feedback loops in predictive policing. In Sorelle A. Friedler and Christo Wilson, editors, *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, volume 81 of *Proceedings of Machine Learning Research*, pages 160–171. PMLR, 2018. URL <https://proceedings.mlr.press/v81/ensign18a.html>.
- [10] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, et al. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2020.
- [11] Charles A. E. Goodhart. Problems of monetary management: The u.k. experience. In *Monetary Theory and Practice: The U.K. Experience*, chapter 4, pages 91–121. Macmillan, London, 1984. doi: 10.1007/978-1-349-17295-5_4. URL https://doi.org/10.1007/978-1-349-17295-5_4.
- [12] Thomas Grote, Konstantin Genin, and Emily Sullivan. Reliability in machine learning. *Philosophy Compass*, 19(5):e12974, 2024. doi: 10.1111/phc3.12974. URL <https://doi.org/10.1111/phc3.12974>.
- [13] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of NAACL*, 2018.

- [14] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. URL <https://arxiv.org/abs/1610.02413>.
- [15] Yili Hong, Jiayi Lian, Li Xu, Jie Min, Yueyao Wang, Laura J. Freeman, and Xinwei Deng. Statistical perspectives on reliability of artificial intelligence systems. *Quality Engineering*, 35(1):56–78, 2023. doi: 10.1080/08982112.2022.2089854. URL <https://doi.org/10.1080/08982112.2022.2089854>.
- [16] Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, New York, 2015. ISBN 978-0-521-88588-1. doi: 10.1017/CBO9781139025751. URL <https://doi.org/10.1017/CBO9781139025751>.
- [17] Robin Jia and Percy Liang. Adversarial examples for evaluating reading comprehension systems. In *Proceedings of EMNLP*, 2017.
- [18] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Proceedings of the Innovations in Theoretical Computer Science Conference (ITCS)*, 2017. URL <https://arxiv.org/abs/1609.05807>.
- [19] Angeliki Lazaridou et al. Mind the gap: Assessing temporal generalization in language models. *Proceedings of NeurIPS*, 2021.
- [20] David Lazer, Ryan Kennedy, Gary King, and Alessandro Vespignani. The parable of google flu: traps in big data analysis. *Science*, 343(6176):1203–1205, 2014.
- [21] Percy Liang, Rishi Bommasani, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.
- [22] Zachary C. Lipton. The mythos of model interpretability. *Queue*, 16(3), 2018.
- [23] Charles F. Manski. *Identification for Prediction and Decision*. Harvard University Press, 2007.
- [24] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453, 2019. doi: 10.1126/science.aax2342.
- [25] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. URL <https://arxiv.org/abs/1906.02530>. Also available in NeurIPS 2019 proceedings.

- [26] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009. URL <https://bayes.cs.ucla.edu/BOOK-2K/>.
- [27] Judea Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2009.
- [28] Joaquin Quiñonero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil D. Lawrence, editors. *Dataset Shift in Machine Learning*. MIT Press, 2009. URL <https://dl.acm.org/doi/10.5555/1462129>. ACM record for the MIT Press volume.
- [29] Joaquin Quiñonero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil Lawrence. Dataset shift in machine learning. *MIT Press*, 2009.
- [30] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishal Shankar. Do imagenet classifiers generalize to imagenet? In *Proceedings of ICML*, 2019.
- [31] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* ’19)*, pages 59–68, 2019. doi: 10.1145/3287560.3287598. URL <https://doi.org/10.1145/3287560.3287598>.
- [32] William R. Shadish, Thomas D. Cook, and Donald T. Campbell. *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin, 2002.
- [33] Marilyn Strathern. ‘improving ratings’: audit in the british university system. *European Review*, 5(3):305–321, 1997. doi: 10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4. URL [https://doi.org/10.1002/\(SICI\)1234-981X\(199707\)5:3<305::AID-EURO184>3.0.CO;2-4](https://doi.org/10.1002/(SICI)1234-981X(199707)5:3<305::AID-EURO184>3.0.CO;2-4).
- [34] Antonio Torralba and Alexei Efros. Unbiased look at dataset bias. In *Proceedings of CVPR*, 2011.
- [35] John R Zech, Marcus A Badgeley, Menglei Liu, Anthony B Costa, Joseph J Titano, and Eric Karl Oermann. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. *PLoS medicine*, 15(11):e1002683, 2018.